Inference from Scientific Data - Worksheet 1

Marks shown in square brackets

Question 1: Working from the definitions of the non-central moments μ_k and central moments ν_k given in the lectures, show that for any positive integer n **[3]**

$$\nu_n = \sum_{k=0}^{n} (-1)^k \, {}^nC_k \, \mu_1^k \, \mu_{n-k} \, .$$

By considering the case $n = 2$, show that **[2]**

$$\mathrm{Var}(x) = \left\langle x^2 \right\rangle - \langle x \rangle^2 \, .$$

Question 2: Let $x_1 \sim \mathrm{Gamma}(k_1, \theta)$ be a *Gamma-distributed* random variable. The Gamma distribution has Probability Density Function (PDF)

$$P(x_1; k_1, \theta) = \begin{cases} \frac{x_1^{k_1 - 1} \exp(-x_1/\theta)}{\Gamma(k_1)\theta^{k_1}} & \text{if } x_1 > 0 \\ 0 & \text{if } x_1 \leq 0 \end{cases},$$

where $k_1 > 0$ and $\theta > 0$ are the *shape* and *scale* parameters of the distribution respectively. The gamma function is defined by the integral

$$\Gamma(z) = \int_0^\infty \mathrm{d}t \; t^{z-1} \exp(-t).$$

On the same set of axes, sketch the PDF of the Gamma distribution for the following sets of parameters: $(k_1, \theta) = (1, 1)$, $(2, 1)$, and $(2, 2)$. **[1]**

Data $D = \{x_1, x_2, ..., x_n\}$ are independent and identically drawn from an exponential distribution with uknown rate, λ.
The likelihood for the data is then,

$$P(D|\lambda) = \prod_{i=1}^{n} P(x_i|\lambda),$$

where

$$P(x_i|\lambda) = \lambda \exp\left(-\lambda x_i\right),$$

for $x_i \geq 0$.
Using a Gamma distribution as a prior with form,

$$P(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \, \lambda^{\alpha - 1} \, \exp\left(-\beta\lambda\right),$$

write down the properly normalised posterior probability distribution, $P(\lambda|D)$, in terms of α, β, n, and S, where $S = \sum_{i=1}^{n} x_i$. **[4]**
Explain each step of your solution. **[2]**
Calculate the mean of the posterior probability density function. **[2]**

Question 3: A discrete random variable x takes values ± 1 with equal probability; i.e.

$$P(x = -1) = P(x = +1) = \frac{1}{2}\,.$$

Comment This is an example of a Bernoulli distribution. *The Bernoulli distribution is a special case of the binomial distribution where a single trial ($n = 1$) is conducted. In this case the probability of successful outcome is one half ($p = 1/2$).*

A new random variable is defined as the sum $z_n = \sum_{i=1}^{n} x_i$, where the x_i are all independently and identically distributed according to the Bernoulli distribution. List the possible values of z_n and find the probability mass function (PMF) for z_n. **[3]**

When $n = 10^6$, find an approximate value for the probability that $z_n = 0$. *Hint: think carefully about possible sources of numerical error. Consider using the central limit theorem.* **[3]**